

GEN AI ASSIGNMENT 2

1. Introduction

Handwritten digit recognition is an important problem in computer vision and deep learning. It is widely used in applications such as postal code recognition, bank cheque processing, and digitizing handwritten documents. In this assignment, a Convolutional Neural Network (CNN) was developed to classify handwritten digits from the MNIST dataset.

The objective of this project is to:

- Load and preprocess the MNIST dataset
- Build a CNN model for digit classification
- Train and evaluate the model
- Test the model on custom handwritten digit images
- Analyze the performance of the model

2. Dataset Description

The MNIST dataset is a standard benchmark dataset in deep learning provided by TensorFlow/Keras.

Dataset characteristics:

Property	Value
Training samples	60,000
Testing samples	10,000
Image size	28 × 28 pixels
Number of classes	10 (digits 0–9)
Image type	Grayscale

Each image contains a handwritten digit labeled from 0 to 9.

3. Data Preprocessing

Data preprocessing is an important step to ensure the data is suitable for training the CNN model.

The following preprocessing steps were performed:

Normalization

Pixel values originally ranged from 0 to 255. These values were normalized to a range between 0 and 1 by dividing by 255. This improves model convergence.

Reshaping

Images were reshaped from:

(28,28)

to:

(28,28,1)

This was necessary because CNN models expect a channel dimension.

One-Hot Encoding

Labels were converted into categorical format using one-hot encoding.

Example:

Digit 3 → [0 0 0 1 0 0 0 0 0]

This allows the model to perform multi-class classification.

4. CNN Model Architecture

A Convolutional Neural Network was designed using multiple layers for feature extraction and classification.

The architecture includes:

- Convolution layer (feature extraction)
- Max pooling layer (dimensionality reduction)
- Dropout layer (prevent overfitting)
- Dense layer (classification)
- Softmax output layer

Model Structure:

Layer	Purpose
Conv2D (32 filters)	Feature extraction
MaxPooling	Reduce dimensions
Conv2D (64 filters)	Deeper features
MaxPooling	Reduce size
Dropout	Prevent overfitting

Layer	Purpose
Flatten	Convert to vector
Dense (128 neurons)	Learning patterns
Dropout	Regularization
Dense (10 neurons)	Output layer

The softmax activation function was used for multi-class classification.

5. Model Training

The model was compiled using:

- Optimizer: Adam
- Loss function: Categorical Crossentropy
- Metric: Accuracy

Training parameters:

Parameter	Value
Epochs	8
Batch size	128

Validation split 10%

During training:

- Training accuracy increased steadily
 - Validation loss decreased
 - Model converged successfully
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6. Model Evaluation

The trained model was evaluated on the test dataset.

Results obtained:

Metric	Result
Test Accuracy	98%+
Test Loss	Very low

The model successfully met the required accuracy threshold.

This shows the CNN model generalized well to unseen data.

7. Custom Image Testing

The trained model was tested using custom handwritten digit images.

Preprocessing steps applied to custom images:

- Converted to grayscale
- Resized to 28×28 pixels
- Normalized pixel values
- Reshaped to CNN format
- Color inversion (to match MNIST format)

Conclusion

In this assignment, a CNN model was successfully implemented for handwritten digit classification using the MNIST dataset. The dataset was properly preprocessed through normalization, reshaping, and encoding. A well-structured CNN architecture was developed and trained successfully.

The model achieved more than 98% accuracy on the test dataset, satisfying the assignment requirements. Testing on custom handwritten images showed that the model can generalize well to new data when proper preprocessing is applied.

Overall, this project demonstrates the effectiveness of CNNs for image classification problems.